Snowpack and runoff

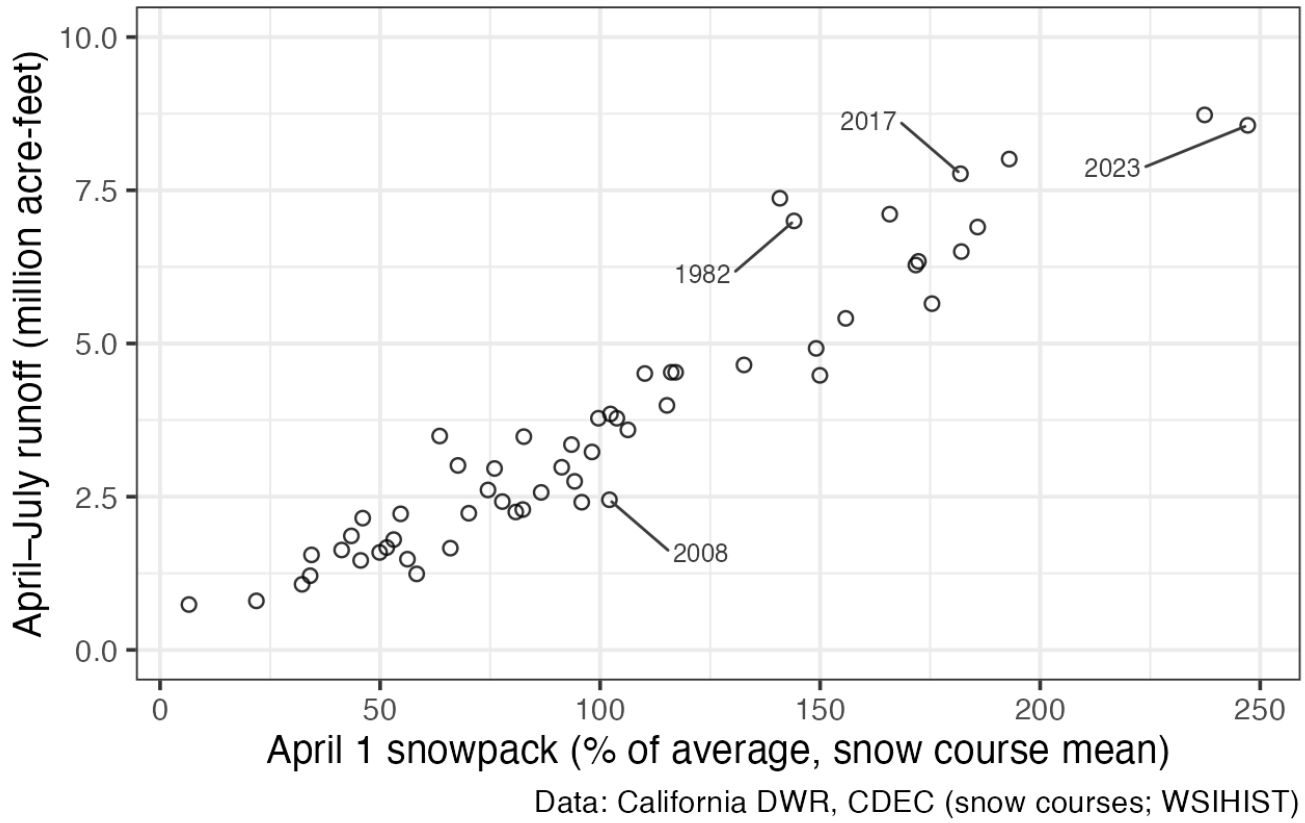
Spring snowmelt is the main source of irrigation water for San Joaquin Valley farms. When runoff is low, water deliveries shrink and farmers pump more groundwater. Wet years refill the aquifer, but long-term overpumping has reduced water tables. Warmer winters mean less snow, so the valley may depend even more on groundwater.

Data

In the following figure, each point represents one year of California's history.

On the x-axis is the year's snowpack index. The snowpack index is a percent of average: each snow course's April 1 snow water equivalent is divided by that same site's long-term April 1 average. The index then averages those percentages across every snow course in the Stanislaus, Tuolumne, Merced and San Joaquin basins, so 100% is a typical year, 150% is a big snow year, and 50% is half the usual snowpack.

The y-axis shows the annual April–July runoff for the San Joaquin Valley, in millions of acre-feet. It's the total spring and summer flow of the Stanislaus, Tuolumne, Merced and San Joaquin rivers.



Model

Below is the summary of a linear regression model, fit to the data above in R.

```
Call:
lm(formula = apr_jul ~ snow_pct, data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-1.26420 -0.34642 -0.05823  0.28867  2.18919

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.152594   0.186807  -0.817    0.418
snow_pct     0.037867   0.001622  23.341   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6585 on 54 degrees of freedom
Multiple R-squared:  0.9098,     Adjusted R-squared:  0.9082
F-statistic: 544.8 on 1 and 54 DF,  p-value: < 2.2e-16
```

In the model summary above, circle the numbers corresponding to the estimates for β_0 and β_1.

Draw the regression line on the figure.

What are the *residuals* for the years 1982, 2008, 2017, and 2023?

Recall a linear regression model in statistical notation is:

$$\hat{y} = \beta_0 + \beta_1 x + \epsilon$$

$$\epsilon \sim Normal(0, \sigma)$$

In other words, a linear regression assumes *normally distributed* residuals. In a normal distribution, 95% of values fall within 2σ of the mean. Eyeball the distribution of all the residuals around the regression line - what is a reasonable estimate for σ?

Carbon costs of energy

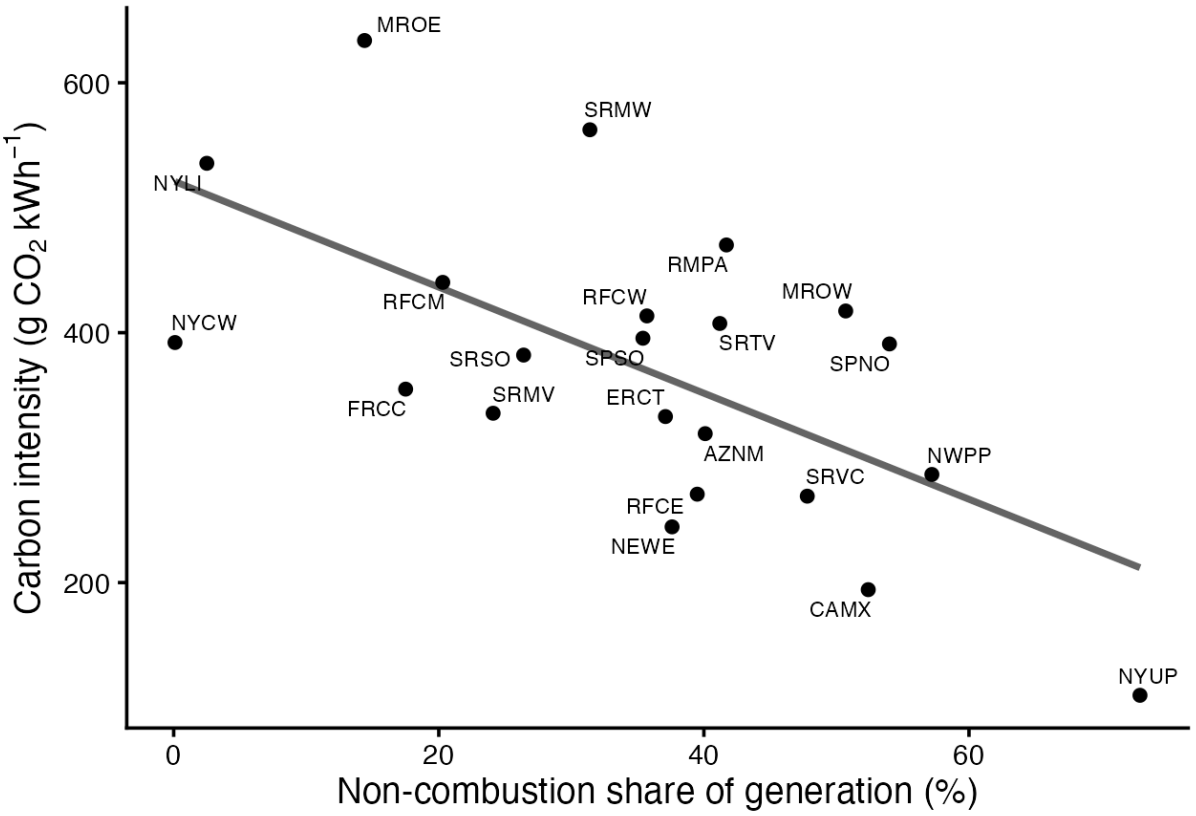
Many climate strategies rely on electrifying everyday activities, such as replacing gas cars with EVs. But an electric car is only as clean as the grid that charges it, and US energy comes from a mix of sources. Policymakers want to know how regional CO_2 emissions respond to shifts in generation toward "non-combustive" sources, such as wind and solar.

Data

In the following figure, each point represents one regional US power grid in 2023. The EPA divides the contiguous United States into 22 subregions, such as CAMX for most of California and ERCT for most of Texas.

On the x-axis is the subregion's non-combustion share of generation. This is the percent of all the electricity the region's power plants generated over the year that came from sources that don't burn fossil fuel: wind, solar, hydro, nuclear and geothermal. So 0% is a grid powered entirely by coal, gas and oil and 50% is a grid where half the electricity comes from non-fossil fuel sources,.

The y-axis shows the subregion's *carbon intensity*, in grams of CO_2 per kilowatt-hour (g CO_2/kWh). It's the total CO_2 emitted by every power plant in the region over the year, divided by the total electricity those plants generated. For scale, a typical coal plant emits roughly 1,000 g CO_2/kWh, and a modern natural gas plant emits roughly 400.



Model

Below is a redacted summary of a linear regression model, fit to the data above in R.

```
Call:
lm(formula = co2_g_kwh ~ low_carbon_pct, data = energy)

Residuals:
    Min      1Q  Median      3Q     Max
-128.77  -83.36  -11.05   56.57  174.32

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     _______     46.758  11.149 4.93e-10 ***
low_carbon_pct  _______      1.186  -3.579  0.00188 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 96.07 on 20 degrees of freedom
Multiple R-squared:  ______,     Adjusted R-squared:   ______
F-statistic: 12.81 on 1 and 20 DF,  p-value: 0.001876
```

What is the equation of the regression line in the figure above?

Fill in the coefficient estimate blanks in the model summary above.

Based on the distribution of the residuals around the regression line, what is a reasonable estimate of σ?

The R^2 value has been removed. How do you think it compares to the R^2 of the snowpack-runoff model?